

PlantMR: an auditable summary-statistics toolkit for plant and crop Mendelian randomization with environment-aware effect heterogeneity

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Abstract

Background

Plant Mendelian randomization (MR) can connect molecular traits to agronomic phenotypes, but plant reference assemblies, environmental strata, linkage disequilibrium (LD) and sample overlap are often implicit. We developed PlantMR 1.1 as a local, open-source toolkit for auditable plant and crop summary-statistics MR with a narrowly defined environment-effect-heterogeneity analysis.

Results

PlantMR validates plant metadata and summary statistics, harmonizes alleles, enforces complete SNP-by-environment instrument grids, accepts signed SNP-LD and environment-correlation matrices, and reports rank-aware heterogeneity diagnostics. In 500-replicate simulations, the covariance-aware estimator had a null rejection rate of 0.054 and 95% coverage of 0.946; for a true environment slope of 0.25, the mean estimate was 0.2498, bias -0.00018 , coverage 0.952 and power 1.00. In an LD stress test with AR(1) LD correlation $\rho=0.6$, correct covariance modeling gave rejection 0.044 and coverage 0.956, whereas an independence approximation gave 0.126 and 0.874. A real Arabidopsis data-contract case retained 48 LD-correlated instruments for AT1G11560 and flowering time at 10°C and 16°C. The primary slope was -0.0371 (SE 0.1562, $P=0.812$), with residual heterogeneity $Q=192.89$, $df=60$, $P=7.33 \times 10^{-16}$.

Conclusions

PlantMR provides a reproducible plant-focused implementation and audit workflow. The simulations support covariance-aware calibration under the tested scenarios, but the Arabidopsis case is a software demonstration rather than independent causal validation. PlantMR does not replace colocalization, formal interaction-MR methods, mixed-model GWAS or functional validation.

Keywords: Mendelian randomization; plant genomics; crop genomics; genotype–environment interaction; eQTL; linkage disequilibrium; summary statistics; reproducibility

Background

Mendelian randomization uses genetic associations as instrumental variables to estimate genetically proxied exposure effects on outcomes.[1,2] The relevance, exchangeability and exclusion-restriction assumptions provide the foundation for interpretation, but their implementation depends on the study population, phenotype, molecular exposure and data-generating design. Summary-statistics MR has enabled large-scale analyses using public genome-wide association studies (GWAS) and platforms such as MR-Base,[10] yet summarized associations are also vulnerable to weak instruments, horizontal pleiotropy, LD and sample overlap.[3–9]

Plant applications add several complications. Natural accessions and breeding panels can exhibit strong geographic structure and kinship. LD decay can differ substantially between species, populations and genomic regions. Expression is strongly dependent on tissue, developmental stage and treatment. Multi-environment trials and stress experiments frequently measure the same genotypes under several conditions, creating correlated estimates rather than independent GWAS datasets. Polyploid dosage, presence/absence variation and structural variation introduce additional representations that cannot be silently converted to ordinary biallelic SNPs; recent Arabidopsis pan-genome work illustrates why local adaptation and non-reference sequence variation should remain explicit.[35]

Existing plant-oriented software demonstrates the value of integrated workflows. MRBGR provides a broad population-scale multi-omics and MR toolbox for maize.[17] The maize drought study that motivates the present work prioritized genes using dynamic eQTLs and a lead-eQTL MR workflow.[18] Plant G×E GWAS

and meta-analysis methods also show that environmental contrasts, population structure and interaction terms must be modeled jointly rather than treated as labels.[25–27] Outside plants, MR-GxE uses gene-by-covariate interactions to detect or correct pleiotropic bias under a constant-pleiotropy assumption,[14] whereas interaction-based MR work distinguishes MR-GxE from MR-GENIUS and emphasizes interaction strength and assumption sensitivity.[15] MR-EILLS targets an invariant causal effect across heterogeneous GWAS summary datasets and evaluates multiple MR and meta-analytic comparators.[16] These methods have different estimands and should not be conflated.

Here we develop PlantMR as a plant/crop summary-statistics software and reporting workflow. The contribution is deliberately narrower than a new general theory of interaction MR: (i) an explicit plant metadata and summary-statistics contract; (ii) complete SNP-by-environment instrument selection; (iii) signed LD and environment-correlation inputs; (iv) rank-aware heterogeneity diagnostics; and (v) a reproducible local implementation with simulations and a real Arabidopsis data-contract case. We evaluate calibration, LD-misspecification risk and failure boundaries rather than claiming a new causal gene discovery.

Implementation

Study questions and workflow

We evaluated four prespecified questions: (1) can the software enforce a plant-aware, auditable data contract; (2) does the covariance-aware environment-slope estimator recover known effects under correlated SNP and environment errors; (3) how much can inference deteriorate when LD and environment dependence are incorrectly treated as independent; and (4) can the same audit trail be executed on a real plant data contract without converting a demonstration into a causal-gene claim? The workflow in Figure 1 separates these questions into input provenance, harmonization, instrument selection, covariance specification, estimation and reporting.

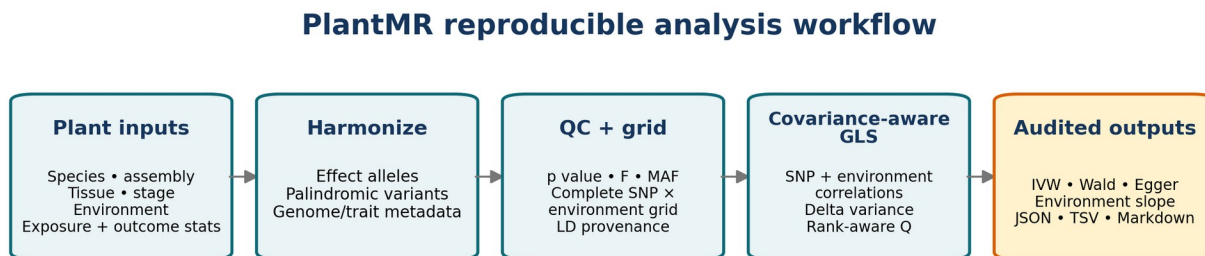


Figure 1. PlantMR reproducible workflow. Plant metadata and summary statistics are retained alongside allele harmonization, complete-grid quality control, covariance specification and machine-readable audit outputs.

PlantMR software contract and analysis workflow

PlantMR accepts exposure and outcome tables containing SNP, effect allele, other allele, beta, standard error and P value. Optional fields include effect-allele frequency and sample size. Plant metadata records species, reference assembly, trait, tissue, developmental stage, environment, ploidy and LD-panel provenance. The software validates numeric values and alleles, harmonizes reversed and complemented alleles, identifies ambiguous palindromic variants, applies P-value/MAF/F-statistic filters and records every exclusion.

Table 1. PlantMR components, boundaries and audit outputs.

Component	Function	Output
Schema and metadata	Validate summary statistics and plant context	Validation messages and metadata JSON
Allele harmonization	Align effect alleles and handle palindromic variants	Harmonized table and exclusion counts
Instrument QC	P value, MAF, F statistic and complete-grid filtering	Instrument audit and retained SNP table
Standard MR	Wald, fixed/random IVW, MR-Egger, Q and leave-one-out	JSON, TSV and Markdown reports
Environment slope	GLS on SNP×environment ratio estimates	Pooled effect, slope, covariance rank and Q
Reproducibility	CLI, tests, simulations, source-data records	Versioned reports and benchmark outputs

The command-line interface exposes `validate`, `run`, `run-stratified` and `run-gxe` commands. The software does not silently substitute SMR, GSMR, colocization, MVMR, MR-PRESSO, MR-GxE, MR-GENIUS or MR-EILLS.

Comparison with related tools

Following the structure used in recent Plant Methods software articles, we compare PlantMR with functionally adjacent tools before presenting the benchmarks. This is a scope and contract comparison based on the cited software papers and public documentation, not a fabricated head-to-head runtime experiment. PlantMR's distinct target is the combination of plant metadata, complete SNP-by-environment grids, signed LD/environment covariance and auditable summary-statistics outputs.

Table 2. Functional positioning against related tools. “Not the primary focus” means that a tool may support a related analysis but does not expose the same contract or estimand.

Tool	Primary scope	Plant context	Environment / LD handling	Evidence used here
PlantMR 1.1	Plant summary-statistics MR and environment-effect heterogeneity	Native metadata, assembly, tissue, stage, ploidy and LD provenance	Signed LD and optional environment covariance; complete grid required	Simulations, LD stress and Arabidopsis case
MRBGR [17]	Population-scale multi-omics, GWAS and MR toolbox	Maize and rice case studies; broad multi-omics workflow	Different MR/network estimands; not the present G×E covariance contract	Literature scope comparison
MR-Base / TwoSampleMR [10]	Human GWAS repository and two-sample MR automation	Human-oriented public GWAS ecosystem	Standard MR sensitivity analyses; plant metadata not the primary contract	Literature scope comparison
metaGE [25]	Multi-environment GWAS meta-analysis	Plant METs and G×E QTL detection	Models heterogeneity and environmental covariates at GWAS meta-analysis level	Estimand comparison; not reimplemented
MAPtools [36]	Mapping-by-sequencing and QTL-Seq command-line analysis	Plant-tested, multi-species mapping workflows	Variant/QTL mapping rather than summary-statistics MR covariance	Workflow/software-article comparison

Environment-effect-heterogeneity model

For SNP j and environment k , the ratio estimate is $r_{jk} = \beta_{y,jk} / \beta_{x,jk}$. Let z_k be a prespecified numeric environmental score. PlantMR fits $r_{jk} = \theta_0 + \theta_1 z_k + \epsilon_{jk}$. θ_0 is the genetically proxied effect at $z=0$ and θ_1 is the change in the MR effect per unit increase in z . This is an effect-heterogeneity model. Its intercept is not the MR-GxE pleiotropy intercept and is not interpreted as a correction for horizontal pleiotropy.

The first-order delta variance is $v_{jk} = \text{se}_{y,jk}^2 / \beta_{x,jk}^2 + \beta_{y,jk}^2 \text{se}_{x,jk}^2 / \beta_{x,jk}^4$. With SNP-major/environment-major ordering, optional signed SNP and environment correlation matrices define $V = D(R_{\text{SNP}} \otimes R_{\text{ENV}})D$, where D is the diagonal matrix of ratio standard errors. The GLS estimator is $\hat{\theta} = (X'V^{-1}X)^{-1}X'V^{-1}r$, with $X=[1,z]$. If V is rank-deficient, residual heterogeneity degrees of freedom are $\text{rank}(V) - \text{rank}(X)$.

The model requires a complete environment grid for each SNP and retains only SNPs that pass exposure P value, F statistic and MAF requirements in every environment. This prevents an apparent environmental slope from being caused solely by changing instrument composition. Missing environment correlation is not treated as verified independence; the report states that a diagonal approximation was used.

Implementation details

The study was organized as a software/methods evaluation. We did not use the real-data case to select a favorable estimator, define a new gene-level claim or tune simulation truth. The main claims concern input validation, covariance-aware estimation and diagnostic transparency.

Input schema and harmonization

Required summary columns were SNP, effect_allele, other_allele, beta, se and pval. Optional columns were eaf and n. The schema rejects empty/duplicate SNP identifiers, non-finite values, non-positive SE, invalid P values, invalid allele characters and equal effect/other alleles. Harmonization recognizes aligned, reversed, complemented and complemented-reversed alleles and uses allele frequencies to retain only resolvable palindromic variants.

Instrument selection

The primary thresholds were exposure $P \leq 5 \times 10^{-8}$, $F \geq 10$ and $MAF \geq 0.05$ in the Arabidopsis case. The GxE selector requires every retained SNP to pass these filters in every environment. This prevents environment-specific tool composition from masquerading as effect heterogeneity. LD is represented by signed correlations, not r^2 values.

Estimators and diagnostics

Standard methods include Wald ratio, fixed-effects IVW, random-effects IVW, MR-Egger, Cochran Q and leave-one-out analysis. The environment-slope estimator uses the delta-method ratio variance and GLS covariance described above. If the covariance is singular, a Moore–Penrose inverse is used for the quadratic form and Q degrees of freedom are rank adjusted.

Simulation design

All simulations used fixed seeds recorded in metadata JSON files. The base benchmark used 20 SNPs and four environment scores (−1.5, −0.5, 0.5, 1.5). Exposure effects were generated as positive strong instruments. Outcome effects were generated from a pooled effect plus a known environment slope, with multivariate normal sampling errors. The LD stress benchmark additionally used an AR(1) LD correlation matrix with $\rho=0.6$.

Arabidopsis data processing

The GSE80744 normalized matrix was downloaded from GEO and the AT1G11560 row extracted. The 1001 Genomes HDF5 matrix was used to extract the local region; five PCs were calculated from every 1000th marker. A public accession VCF supplied REF/ALT labels. AraPheno FT10 and FT16 values were merged by accession ID. Local association regressions included genotype and PC1–PC5. Source URLs, hashes, generated tables and limitations are recorded with the case.

Statistical reporting

All numerical results are reported with effect estimates, standard errors, P values, instrument counts, covariance source and heterogeneity diagnostics, following the reporting principles of STROBE-MR.[13] No genome-wide multiple-testing claim is made for the single-gene Arabidopsis demonstration. Any future genome-wide run must prespecify gene-level aggregation, multiple-testing control and an independent validation set.

Results

Simulation calibration and LD stress

The primary simulation benchmark used 20 instruments, four environments, environmental correlation 0.5, exposure SE 0.01, outcome SE 0.04 and 500 replicates per scenario. It included a null slope, a true slope of 0.25 and a common directional-pleiotropic component. The LD stress benchmark used the same dimensions with AR(1) signed LD correlation $\rho=0.6$ and compared correct covariance specification with a diagonal independence approximation.

Table 3. Primary simulation calibration and LD-stress summary. Dashed lines in Figures 2 and 3 mark nominal 0.05 rejection and 0.95 coverage targets where applicable.

Scenario	Covariance	Mean/bias	Coverage	Rejection or power
Null	Environment-aware	slope bias -0.00069	0.946	0.054
Null	Diagonal	slope bias -0.00071	0.994	0.006
Causal G×E	Environment-aware	mean 0.24982; bias -0.00018	0.952	1.000
Causal G×E	Diagonal	mean 0.24832; bias -0.00168	1.000	1.000
LD stress null	Correct LD+environment	bias 0.00172	0.956	0.044
LD stress null	Diagonal misspecified	bias 0.00177	0.874	0.126
LD stress causal	Correct LD+environment	bias 0.00136	0.970	1.000
LD stress causal	Diagonal misspecified	bias 0.00170	0.880	1.000

The directional-pleiotropy scenario exposed an important limitation. The environment slope remained approximately unbiased when the true slope was zero, but the pooled intercept was biased by +0.216. Therefore, a stable environment slope cannot be interpreted as proof that the pooled causal relation is protected from horizontal pleiotropy.

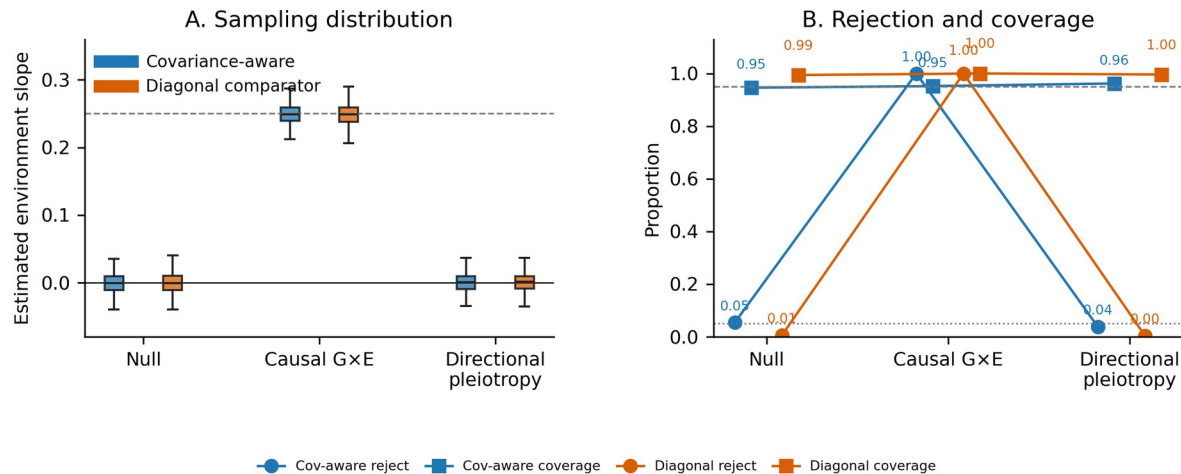


Figure 2. Primary simulation benchmark. The panels summarize slope estimation, rejection behavior and 95% confidence-interval coverage across null, causal environment-slope and directional-pleiotropy scenarios.

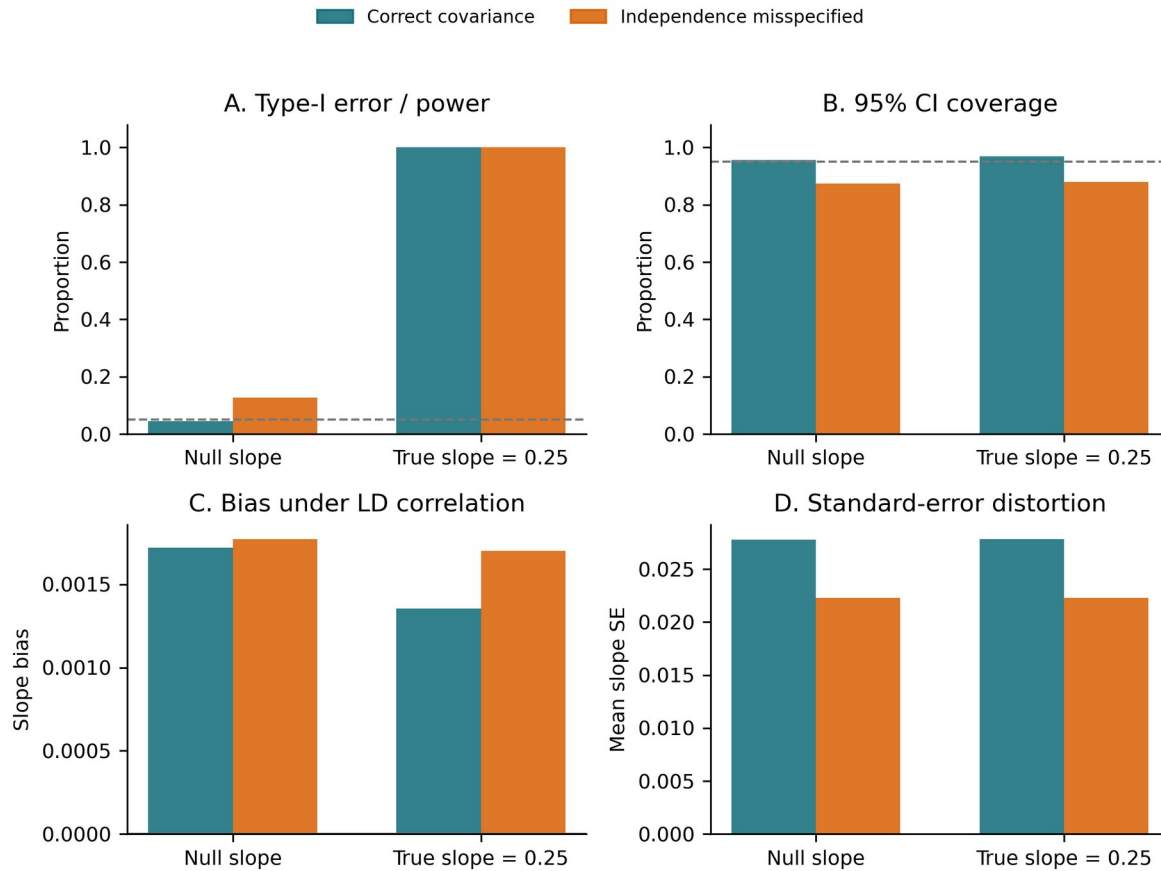


Figure 3. LD stress benchmark. Treating correlated SNP and environment errors as independent preserved approximate point estimates but produced smaller standard errors, inflated null rejection from 0.044 to 0.126 and reduced coverage from 0.956 to 0.874.

Shared-input external-method comparison

We implemented a narrow summary-data MR-GxE comparator following the three-step construction of Spiller et al.[14]. The comparator forms one fixed weighted allele score from exposure associations only within each environment, obtains score–exposure and score–outcome associations, and regresses the latter on the former with an intercept. The same 20 SNPs, four environment strata, environment-correlation matrix, standard errors and 500 replicates per scenario were passed to both methods. The comparison is method-specific rather than a single performance leaderboard: PlantMR estimates an environment-effect slope, whereas summary MR-GxE estimates an invariant causal effect and a constant-pleiotropy intercept.

Table 4. Shared-input PlantMR versus summary-data MR-GxE benchmark. Bias and coverage are reported only where the method’s estimand is defined by the simulated data-generating model; out-of-target outputs are retained as diagnostics.

Scenario	Method / target	Mean estimate	Bias	Coverage	Rejection	Interpretation
Null causal + pleiotropy	Summary MR-GxE / causal effect	−0.0021	−0.0021	0.934	0.066	Target-defined causal calibration
Null causal + pleiotropy	PlantMR / environment slope	−0.0692	—	—	—	Diagnostic only: pleiotropy makes slope non-causal
Constant effect, no pleiotropy	PlantMR / environment slope	0.0036	0.0036	0.938	0.062	Target-defined heterogeneity calibration
Constant effect, no pleiotropy	Summary MR-GxE / causal effect	0.5003	0.0003	0.958	1.000	Target-defined causal calibration
Constant effect + pleiotropy	Summary MR-GxE / causal effect	0.4983	−0.0017	0.938	1.000	Correct causal target under constant pleiotropy

Constant effect + pleiotropy	PlantMR / environment slope	-0.0653	—	—	—	Diagnostic only: not a causal slope
Environment-effect heterogeneity	PlantMR / environment slope	0.2515	0.0015	0.942	1.000	Correct environment-slope target
Environment-effect heterogeneity	Summary MR-GxE / causal effect	1.2143	—	—	—	Diagnostic only: invariant-effect target not defined

The comparison supports complementarity rather than a winner. Under constant causal effect with directional pleiotropy, summary MR-GxE recovered the causal effect (mean 0.4983; coverage 0.938), while the PlantMR environment slope was a diagnostic signal caused by the violated exclusion restriction. Under true environment-effect heterogeneity without pleiotropy, PlantMR recovered the slope (mean 0.2515; coverage 0.942), while the summary MR-GxE output does not have the same causal estimand. The external comparator is therefore evidence about scope and assumptions, not a claim that one method dominates the other.

Shared-input comparison: PlantMR and summary-data MR-GxE

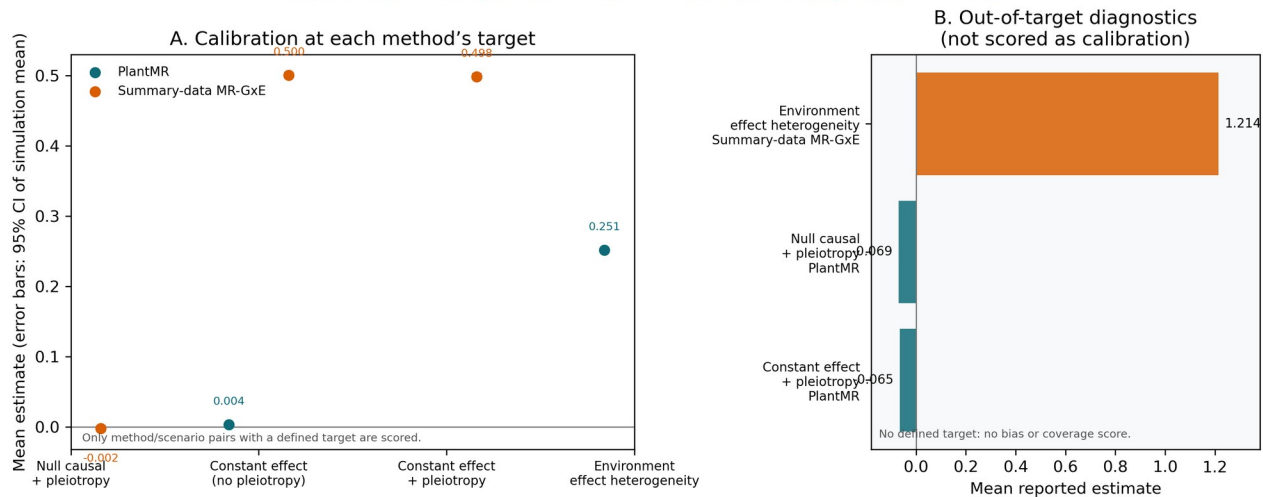


Figure 5. Shared-input comparison. Panel A scores only method/scenario pairs with a defined target. Panel B retains out-of-target estimates as diagnostics and explicitly excludes them from bias and coverage scoring.

Arabidopsis data-contract case

The Arabidopsis case was designed as a reproducibility and diagnostics demonstration rather than a new causal discovery. Baseline leaf expression of AT1G11560 was extracted from the GSE80744 normalized expression matrix, local genotypes were taken from the 1001 Genomes v3.1 matrix and flowering time at 10°C and 16°C was taken from AraPheno. The local association model used ordinary least squares with five whole-genome PCs, following the general principle that population structure and kinship must be modeled in plant association analyses,[29–34] but it was not presented as a reimplement of the published mixed-model/SMR analysis.[19–24]

Table 5. Arabidopsis data-contract inputs and local processing.

Layer	Public source	Local processing
Molecular exposure	GSE80744 normalized expression	AT1G11560 row; log1p transformation; baseline exposure
Genotype	1001 Genomes v3.1	Chr1:3,861,124–3,901,085; 3,352 local SNPs; five PCs
Outcome environment 1	AraPheno FT10	10°C flowering time; z=-3
Outcome environment 2	AraPheno FT16	16°C flowering time; z=+3
Alleles	1001 Genomes accession VCF	REF/ALT mapping for dosage-coded matrix

At $P \leq 5 \times 10^{-8}$, $F \geq 10$ and $MAF \geq 0.05$, 48 SNPs passed in both environments. They were strongly correlated, so the primary analysis supplied a signed LD correlation matrix. The primary diagonal-environment-

covariance run estimated an intercept of 2.3601 (SE 0.4685, $P=4.72 \times 10^{-7}$) and an environment slope of -0.03705 (SE 0.15617, $P=0.81249$). Residual heterogeneity was $Q=192.89$ on 60 rank-adjusted degrees of freedom ($P=7.33 \times 10^{-16}$).

Table 6. Instrument audit for the Arabidopsis case. Exclusion counts are overlapping diagnostics and must not be added as if they were mutually exclusive.

Audit stage	Count	Interpretation
Raw complete SNP×environment grid	172 SNPs / 344 rows	Two environments were available for every raw local SNP.
Ambiguous palindromic exclusions	3 per environment	Removed during allele harmonization; 169 SNPs / 338 rows remained.
P-value diagnostic exclusions	121 rows	Overlapping with other QC exclusions; not a unique-stage count.
F-statistic diagnostic exclusions	106 rows	Overlapping with P-value and complete-grid diagnostics.
Final complete-grid instruments	48 SNPs / 96 rows	Passed the prespecified P-value, F and MAF requirements in both environments.
LD matrix / covariance rank	48×48 LD; rank(V)=62	Rank-aware Q used 62–2=60 residual degrees of freedom.

Table 7. Environment-stratified comparator estimates. Estimates are local OLS summary-statistics demonstrations and are not independent validation of the causal model.

Environment	Estimator	Estimate	SE	P value	Q (P value)
10°C	Fixed IVW	6.6001	0.2038	4.999×10^{-230}	84.90 (5.89×10^{-4})
10°C	Random IVW	7.0842	0.2970	1.030×10^{-125}	84.90 (5.89×10^{-4})
10°C	MR-Egger	4.2259	0.5188	3.758×10^{-16}	—
16°C	Fixed IVW	8.2379	0.2933	1.402×10^{-173}	97.70 (2.05×10^{-5})
16°C	Random IVW	8.8667	0.4561	3.598×10^{-84}	97.70 (2.05×10^{-5})
16°C	MR-Egger	6.7552	0.7683	1.466×10^{-18}	—

The two environment-stratified IVW estimates are both positive but differ in magnitude (fixed IVW 6.6001 at 10°C versus 8.2379 at 16°C). Their heterogeneity statistics are significant, which motivates—but does not by itself identify—the environment-slope analysis. MR-Egger estimates are lower and have large intercept diagnostics (10°C intercept 2.9281, $P=3.59 \times 10^{-8}$; 16°C intercept 2.0160, $P=0.0103$), consistent with the need to treat pleiotropy as an unresolved limitation rather than a solved problem.

A sensitivity run used the Pearson correlation between FT10 and FT16 across 1,122 shared AraPheno accessions ($r=0.88195$) as a phenotype-correlation proxy. It estimated an intercept of 1.8027 (SE 0.6217, $P=0.00374$) and slope -0.08228 (SE 0.06634, $P=0.21491$). This proxy is not known ratio-error covariance and is not treated as primary inference.

Interpretation

The case shows that PlantMR can make plant data provenance, LD dependence, complete-grid selection and covariance approximations visible. It does not prove AT1G11560 causality, because expression and outcome cohorts share accessions, exposure–outcome covariance is not estimated, local OLS differs from the published LMM and residual heterogeneity remains.

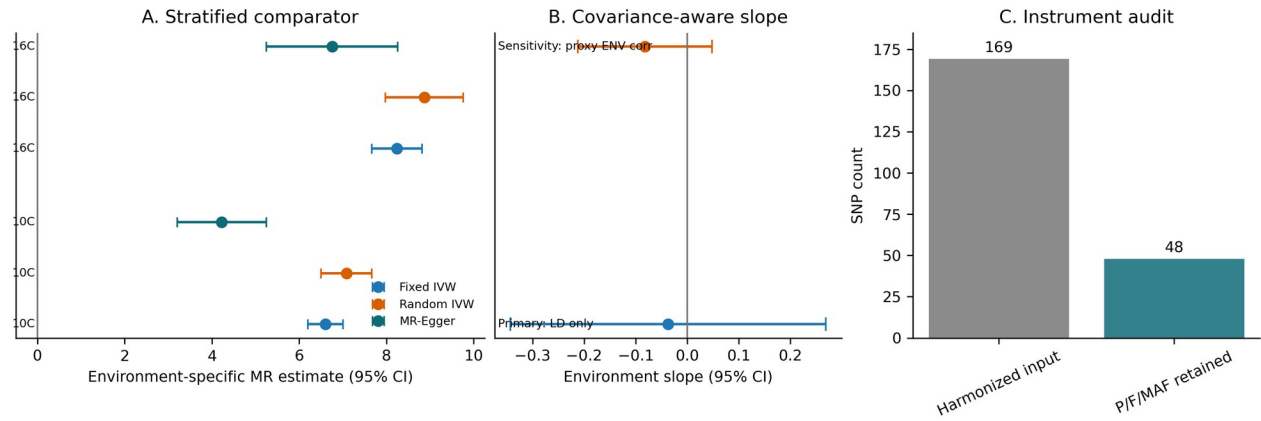


Figure 4. Arabidopsis data-contract case. (A) Environment-stratified comparator estimates; (B) primary and phenotype-correlation-proxy environment slopes; (C) the raw-to-final instrument audit. Error bars are 95% confidence intervals where applicable.

Reproducibility and runtime

The local runtime benchmark completed validation, a synthetic G×E run and the Arabidopsis case with return code 0. The three tasks required 9.70–9.93 seconds and peak resident memory of 126,664–133,424 KB in the frozen environment. These are reproducibility benchmarks on one host, not claims of universal performance or genome-wide scalability.

Table 8. Local runtime benchmark for the frozen PlantMR implementation.

Task	Wall time (s)	Peak RSS (KB)	Return code
validate	9.7034	126,664	0
gxe_synthetic	9.7974	129,736	0
gxe_arabidopsis	9.9300	133,424	0

Discussion

Principal findings

PlantMR addresses a recurring problem in plant summary-statistics MR: genotype, tissue, developmental stage, environment, LD and sample overlap are often stored separately. The workflow links these fields to the estimates and carries them into the report. This is important because the same SNP-level beta and standard error can describe different estimands when the exposure tissue, developmental stage or treatment changes. The contribution is therefore a reproducible data and analysis workflow, not a new claim about plant causality.

The environment-slope analysis answers a narrower question: does the genetically proxied exposure–outcome association change along a prespecified environmental scale? The slope is not a universal gene-by-environment causal parameter and is not a pleiotropy-correction intercept. Its use depends on a common SNP-by-environment instrument grid, an interpretable environmental contrast and a covariance model that matches the data structure.

The simulations show why these details affect inference. Correct and diagonal covariance models gave similar point estimates in the tested settings, but the independence approximation reduced standard errors under LD stress. Null rejection increased from 0.044 to 0.126, and coverage fell from 0.956 to 0.874. The result supports reporting covariance provenance and treating a diagonal analysis as a sensitivity analysis when dependence is plausible.

Comparison with related methods

PlantMR is intended to be used alongside established MR methods. IVW, MR-Egger, robust and median estimators, pleiotropy diagnostics, colocalization and fine-mapping address different sources of uncertainty. [3–12,28] The standard estimators included here provide a common audit trail; they do not make invalid instruments valid. The large Q statistic and non-zero Egger intercepts in the Arabidopsis case are therefore part of the result, not problems that can be removed by choosing another estimator.

The shared-input comparison also separates PlantMR from interaction-based MR. Summary-data MR-GxE uses gene-by-covariate interactions in the instrument–exposure associations and targets an invariant causal effect with a pleiotropy term.[14] MR-GENIUS uses a different identification strategy and requires its own conditions on interaction strength and heterogeneity.[15] MR-EILLS targets an invariant causal effect across heterogeneous GWAS summaries.[16] PlantMR instead estimates change along an observed environmental scale. The four quantities should not be treated as interchangeable.

PlantMR occupies a narrower position than broad plant multi-omics toolboxes. MRBIGR integrates genotype, transcriptome, metabolome, GWAS and MR analyses in maize and demonstrates the workflow with maize and rice data.[17] metaGE addresses multi-environment GWAS meta-analysis and compares fixed-effect, random-effect and alternative meta-analysis procedures across plant datasets.[25] MAPtools emphasizes command-line workflow, published-data case studies and reproducible outputs.[36] PlantMR adopts the same practical sequence, from an explicit input workflow to controlled benchmarks and a plant case, while concentrating on covariance-aware summary-statistics MR.

Simulation and covariance behavior

Plant panels often contain related accessions, local haplotypes and uneven LD. The covariance of ratio estimates therefore cannot be assumed to be diagonal because the input has one row per SNP. Dependence can also arise across environments when accessions are phenotyped repeatedly, trials share controls or environmental measurements are related. In these settings, the covariance model affects standard errors and heterogeneity more directly than the point estimate.

The rank-aware Q calculation provides a direct response to singular or nearly singular covariance matrices. The primary case contained 96 ratio observations, but the supplied covariance had rank 62. The reported residual degrees of freedom were therefore $\text{rank}(V) - \text{rank}(X) = 60$ rather than 94 or 95. This calculation does not verify the biological correctness of the covariance matrix; it prevents the report from treating linearly dependent observations as independent.

The stress benchmark also separates calibration from power. Both covariance choices had power 1.00 for the tested slope of 0.25, although the diagonal model failed the null calibration. A benchmark based on power alone would miss this difference. Null rejection, coverage, bias, RMSE and standard-error behavior should therefore be reported together for covariance-aware methods.

Application to Arabidopsis data

The Arabidopsis analysis was used as a public-data case study of the workflow. The local region provided a complete two-environment grid; ambiguous palindromic variants were removed, and 48 SNPs passed the final instrument filters. The environment-stratified fixed IVW estimates were positive at both temperatures but differed in magnitude. The covariance-aware environment slope was -0.03705 (SE 0.15617 , $P=0.81249$), whereas residual heterogeneity was high ($Q=192.89$, 60 rank-adjusted degrees of freedom, $P=7.33 \times 10^{-16}$).

The case does not establish AT1G11560 as a causal regulator of flowering time. Expression and outcome accessions overlap, the exposure–outcome sampling covariance is unavailable, and the local association model uses ordinary least squares rather than the published mixed model. The baseline expression proxy was also carried across the two outcome environments instead of being estimated as an environment-

specific eQTL. The appropriate interpretation is that PlantMR exposes these dependencies and produces reproducible sensitivity results; the biological conclusion remains provisional.

The phenotype-correlation sensitivity illustrates the same point. The FT10–FT16 correlation changed the fitted intercept and slope, but it is not a measured covariance of the exposure and outcome GWAS estimates. It is therefore useful as a sensitivity calculation, not as a formal correction for sample overlap. Reporting the proxy and its status prevents a convenient correlation from being mistaken for an identified covariance.

Use in plant and crop studies

For natural-accession studies, the report should identify the accession panel, geographic structure strategy, kinship or mixed-model method, reference assembly and allele representation used to construct the LD matrix. For breeding panels and multi-environment trials, the environmental score should be defined before analysis and represent a biological contrast rather than a post hoc label. Measurement error and correlation among temperature, drought, photoperiod or nutrient variables should be documented.

Expression-mediated analyses also require the exposure context to remain attached to the association. Tissue, developmental stage, treatment and normalization should be reported with each eQTL. A baseline eQTL should not be used silently as a stress-specific exposure. Mediation claims require colocalization, fine-mapping or HEIDI-type evidence in addition to MR.[11,12,28] Environment-specific biological questions may require environment-specific exposure models.

Polyploid dosage, homoeolog assignment, subgenome-specific LD, presence–absence variants and structural variants require explicit data models. They are not silently recoded as diploid SNPs in the current implementation. This keeps a software limitation separate from a biological negative result and identifies the inputs that must be defined before those analyses can be interpreted.

Limitations

The Arabidopsis case is not independent causal validation. It uses public processed data, overlapping accessions and a local OLS model. Only two outcome environments are available, so the slope is a contrast along a prespecified scale rather than evidence for linearity, a threshold or a general response curve.

The directional-pleiotropy simulation covers one structured scenario. It does not cover balanced or correlated pleiotropy, weak instruments, nonlinear effects, environmental measurement error, winner's curse or population-stratified LD mismatch. The current software accepts supplied signed correlation matrices, but it does not estimate a causal LD panel, repair every non-positive-semidefinite input or model polyploid dosage and structural variation.

The external comparison is deliberately limited to a summary-data MR-GxE implementation. A full individual-level MR-GxE analysis and independent implementations of MR-GENIUS or MR-EILLS are outside the current study. These methods should be compared only after their input contracts and target estimands have been aligned. Studies that extend PlantMR to independent populations, additional environments, mixed-model summary statistics and functional evidence will be needed before biological claims can be generalized beyond this demonstration.

Conclusions

PlantMR provides a reproducible workflow for plant summary-statistics MR and a defined environment-effect-heterogeneity analysis. Evidence includes explicit metadata, complete-grid instrument selection, covariance-aware GLS, rank-aware heterogeneity reporting, calibrated simulations, a shared-input MR-GxE comparison and a public Arabidopsis case. These results support using PlantMR to make assumptions visible; they do not establish AT1G11560 as a causal gene or replace colocalization, mixed-model GWAS, interaction-MR methods or functional validation.

Availability and requirements

PlantMR is released under the MIT License. The software snapshot is v1.1.0 and the complete manuscript/benchmark/Word-document package is frozen at tag v1.1.0-paper. The public repository is <https://github.com/Zhangzhishuai-HIT/PlantMR>, and the source archive `release/PlantMR\_v1.1.0-paper\_source.zip` is included for submission as supplementary software. A Zenodo or equivalent archival DOI remains to be assigned.

Public data sources include GSE80744 normalized expression data, AraPheno FT10/FT16, 1001 Genomes v3.1 and the Arabidopsis source publications. The large 1001 Genomes provider archive is not redistributed; the derived local genotype region, PC scores, allele mapping and provider checksums are included. The maize supplementary workbook, eQTL table and candidate table are included for audit but not used as a new causal result.

Table 9. Plant Methods software availability and requirements.

Field	Current value
Project name	PlantMR 1.1
Project home page	https://github.com/Zhangzhishuai-HIT/PlantMR
Operating system(s)	Linux verified; platform-independent Python code intended.
Programming language	Python 3.10 or later.
Dependencies	NumPy, pandas, SciPy, statsmodels, matplotlib and psutil; declared in environment.yml.
License	MIT License.
Restrictions for non-academic use	No additional restriction stated in the local license.

List of abbreviations

Abbreviation	Definition
MR	Mendelian randomization
GWAS	Genome-wide association study
eQTL	Expression quantitative trait locus
G×E / GxE	Genotype-by-environment interaction or environment-effect heterogeneity
LD	Linkage disequilibrium
GLS	Generalized least squares
IVW	Inverse-variance weighted
MAF	Minor allele frequency
SNP	Single-nucleotide polymorphism
Q	Cochran-type heterogeneity statistic

Declarations

Declaration	Status
Ethics approval and consent to participate	Not applicable: public plant accessions and public aggregate/processed data were used.
Consent for publication	Not applicable.
Availability of data and materials	Public source data and derived audit materials are described above; code is available at https://github.com/Zhangzhishuai-HIT/PlantMR

	HIT/PlantMR; archival DOI pending.
Competing interests	To be completed by authors.
Funding	To be completed by authors.
Author contributions	To be completed by authors.
Acknowledgements	To be completed by authors or marked Not applicable.
Authors' information	Author names, affiliations and corresponding author to be supplied.

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